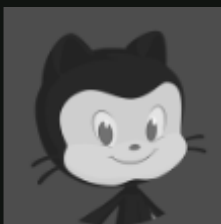


GitHub Agentic Workflows

PUBLIC PREVIEW



MONA LISA

Mascot
GitHub

MADE WITH ♥ BY

GitHub Next, Microsoft Research,
Azure Core Upstream & GitHub Actions

Repos that clean themselves

📁 Repositories

- FSharp.Data
- Deedle
- SwaggerProvider
- FSharp.Formatting
- FSharp.Control.AsyncSeq
- GenPRES
- FSharp.TypeProviders.SDK
- FSharp.Control.TaskSeq
- - licensee
- - fantomas ✓
- - FsAutoComplete ✓
- - dowhy ✓
- fsharp
- FSharp.Stats ✓

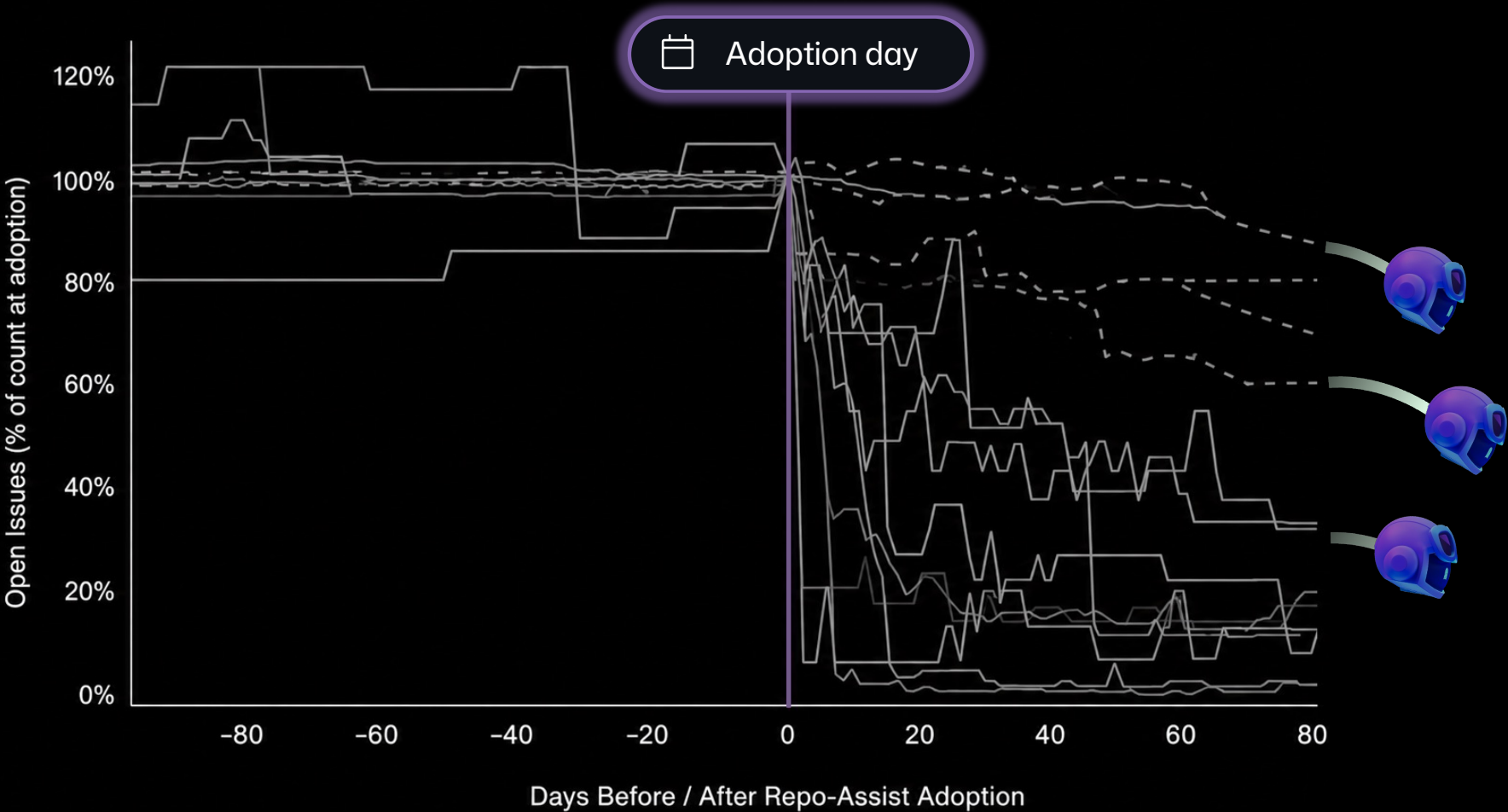
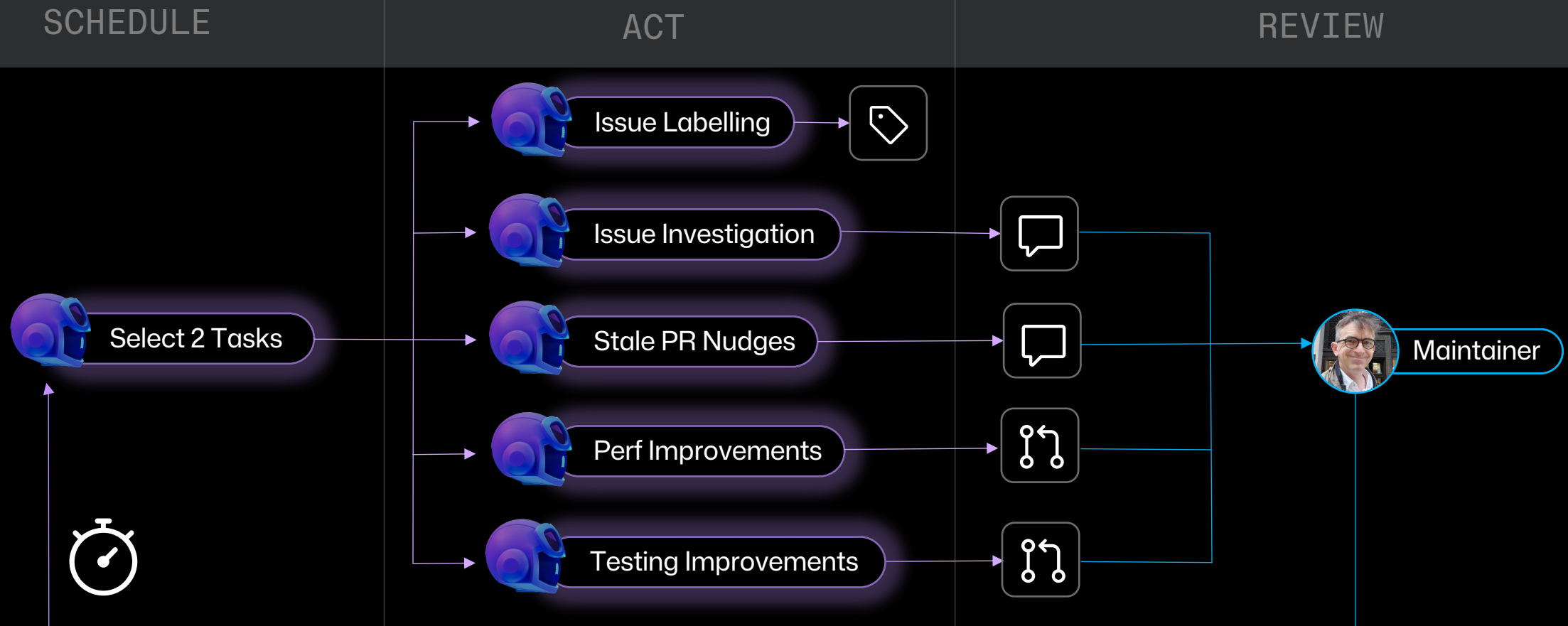


Figure: Burning down open issues in F# repositories with “Repo Assist” workflow.

Repo Assist

Auto Maintenance Loop



github.com

agentics / workflows / repo-assist.md

↑ Top

PreviewCodeBlame423 lines (332 loc) · 25.5 KB

Raw

When

Who

Where

What

Guardrails

Agent

```
on:
  schedule: every 12h
  workflow_dispatch:
    inputs:
      command:
        description: "Optional command-mode instruction (for example: Run Task 9)"
        required: false
        type: string
        default: ""
      slash_command:
        name: repo-assist
        reaction: "eyes"
    permissions:
      pull-requests: read

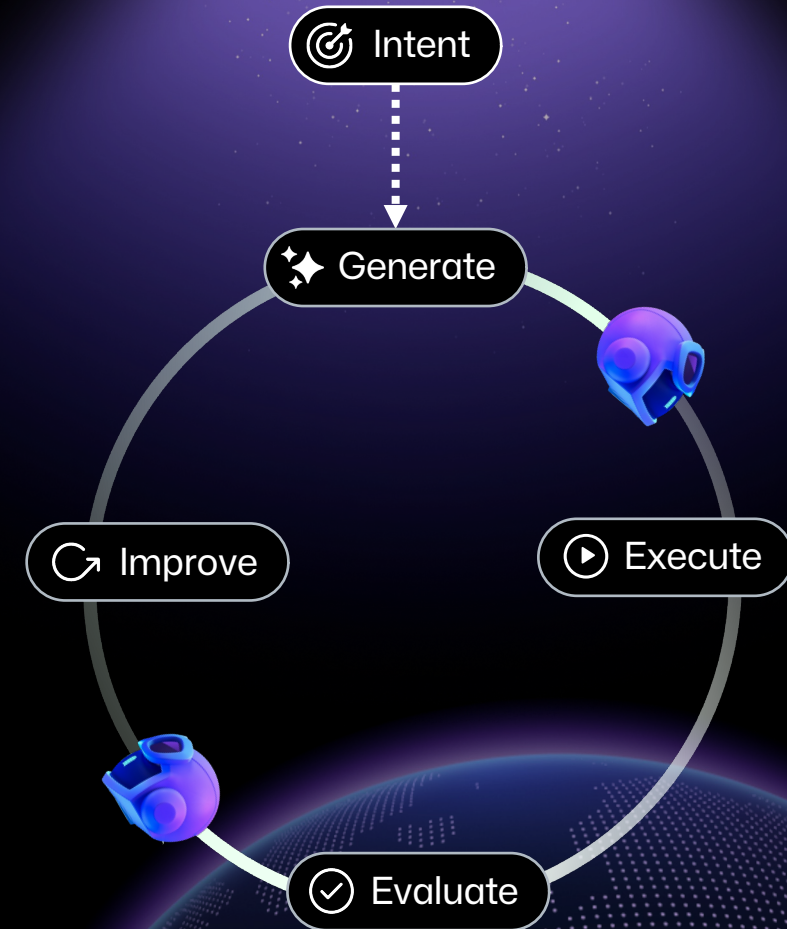
66 safe-outputs:
93   if: github.event.pull-request:
77     draft: true
78     title-prefix: "[repo-assist] "

200 # Repo Assist
201
202 ## Command Mode
203
    Take heed of **instructions**: "${{ steps.sanitized.outputs.text || inputs.command }}"

206 If these are non-empty (not ""), then you have been triggered via `/repo-assist <instructions>` (or by the user setting `inputs.command`)
```

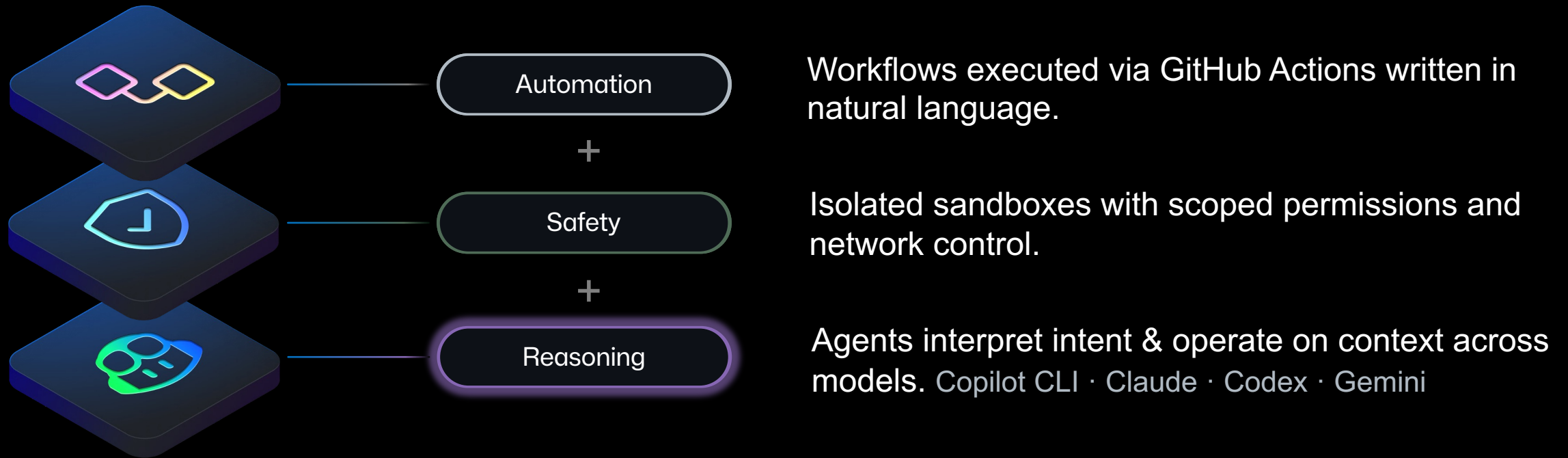

Human Agent Processes

- 🎯 Agents accelerate the entire SDLC.
Every step of our software process can be “agentified”.
- ✅ Automation is the true accelerator.
Completion: 1.4x, Dev solo: 10x, Automation: 100-1000x.
- 🔄 Scale with processes not swarms.
Orchestration not chaos.



GitHub Agentic Workflows

Human Agent Processes powered GitHub.



Orchestrated Automation through Human Agent Processes

issue-clarifier.md ×

.github > workflows > issue-clarifier.md > abc # Issue Clarifier

```
1 ---
2 on:                                # Trigger: when to run
3   issues:
4     types: [opened]
5
6 permissions: read-all             # Security: read-only by default
7
8 safe-outputs:                      # Allowed write operations
9   add-comment: _____
```

 Sandbox

```
10 ---
11 # Issue Clarifier
12
13 Analyze the current issue and ask for additional details
14 if the issue is unclear.
15
```

 Agent

mnkiefier opened 24 minutes ago



Developer

Please add better setup instructions.

Right now it's not clear how to get started, what prerequisites are needed, or what the expected workflow is.

Create sub-issue



github-actions bot 20 minutes ago – with [GitHub Actions](#)

Thanks for raising this! The request for better setup instructions is clear, but to write something accurate and useful, a few more details would help:

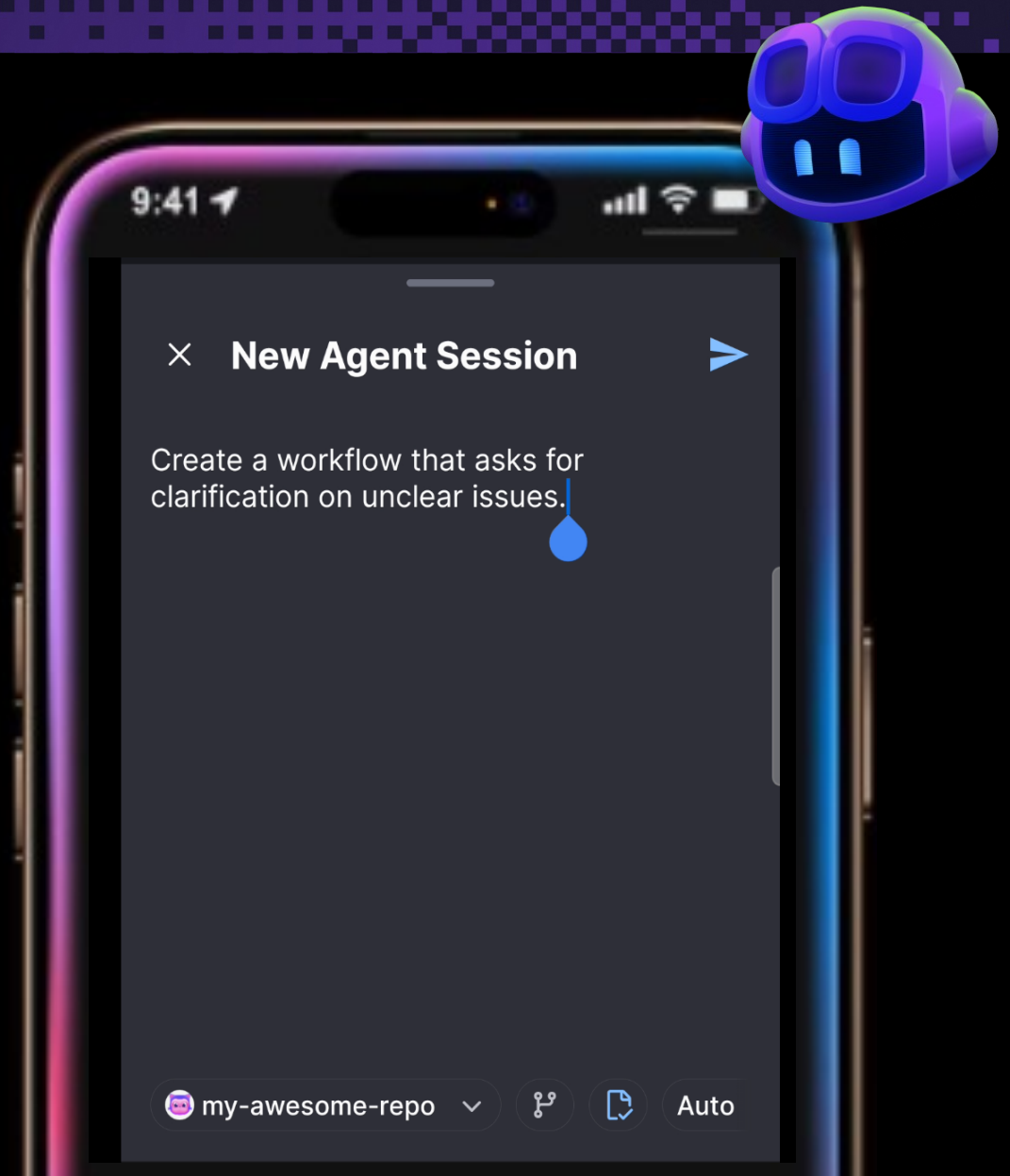
1. **What does this project do?** A brief description of the project's purpose would help frame the setup instructions appropriately.

Spec Driven Automation through Agentic Authoring

Write agents with the help of agents, iteratively.

- ✓ Configure your repository
for GitHub Agentic Workflows
- ✓ Create new workflows
- ✓ Debug existing workflows
- ✓ Optimize Token Usage

☐ Spec Driven Development



 **my-awesome-repo** Private

👁 Watch 0 🍴 Fork 0 ★ Star 0

🔗 main 🔗 2 Branches 🏷 0 Tags

🔍 Go to file t + 🔄 <> Code

 mnkieferr	Merge pull request #2 from mnkieferr/add-workflo...	530e9 · 7 minutes ago	🕒 3 Commits
📁 .github/workflows	Add agentic workflow daily-repo-s...	7 minutes ago	
📄 .gitattributes	Add agentic workflow daily-repo-s...	7 minutes ago	
📄 README.md	Add initial README with project description	minutes ago	

📖 README

Welcome to my-awesome-repo

This is an awesome repo! 🐙

About

No description, website, or topics provided.

- 📖 Readme
- 📈 Activity
- ★ 0 stars
- 👁 0 watching
- 🍴 0 forks

Releases

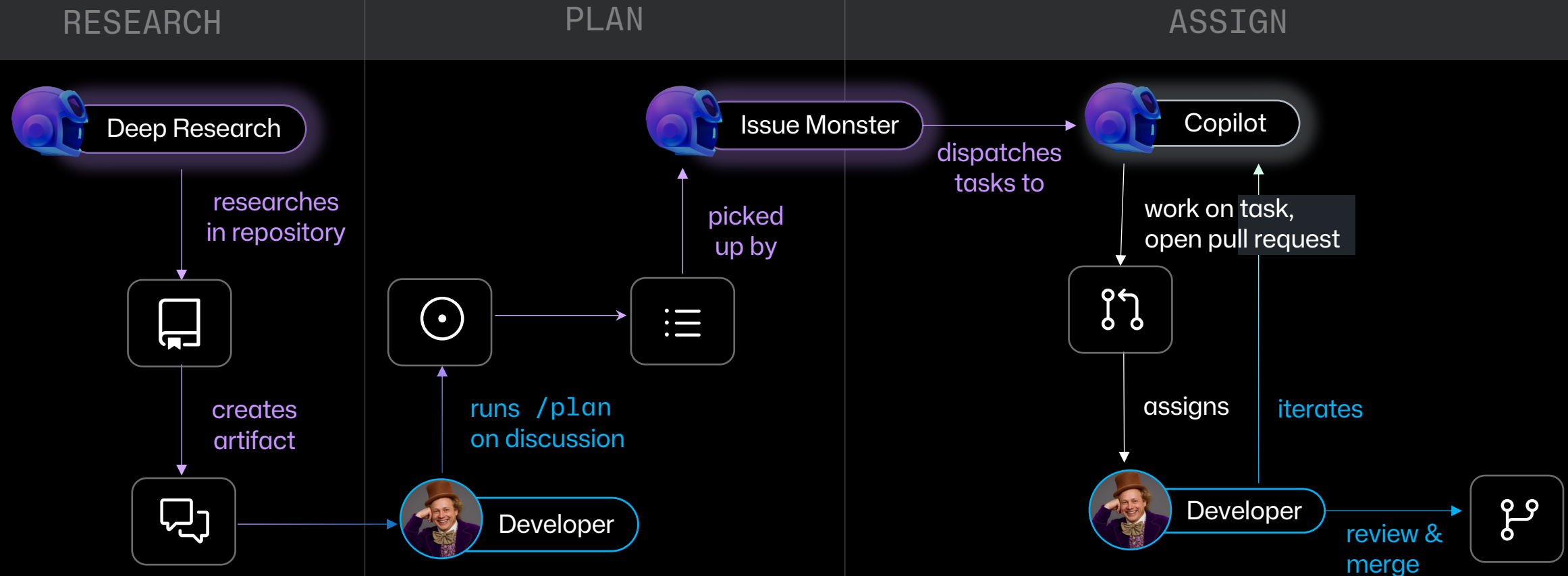
No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

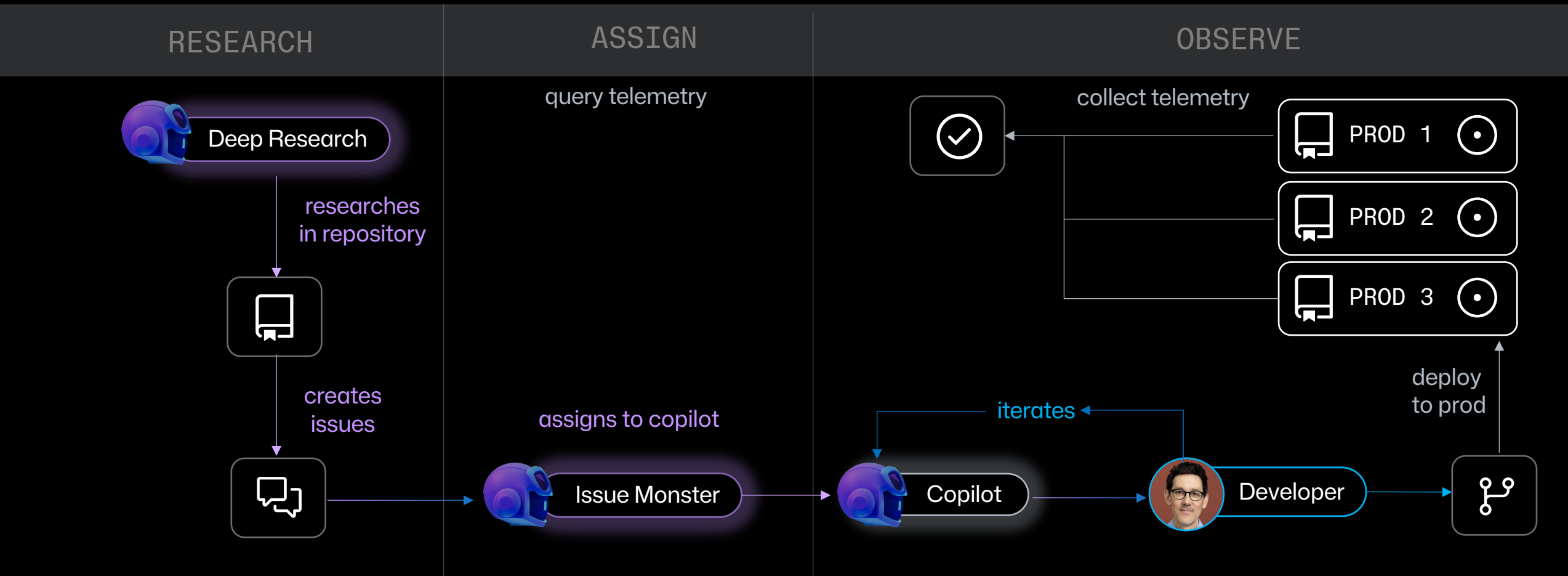
Research – Plan – Assign

Daily Test Improver, Daily Perf Improver, ...



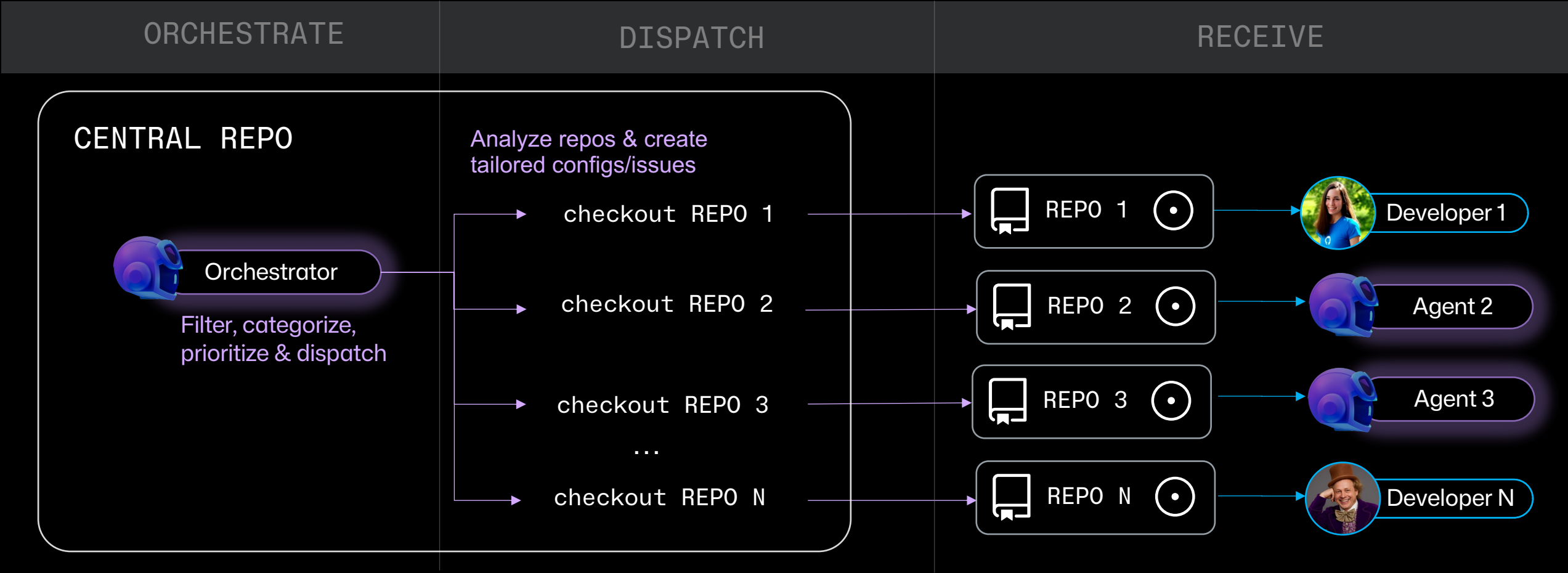
Research – Assign – Observe

Agentic Feedback Loop



Central Repo Ops

Rolling out large scale operations

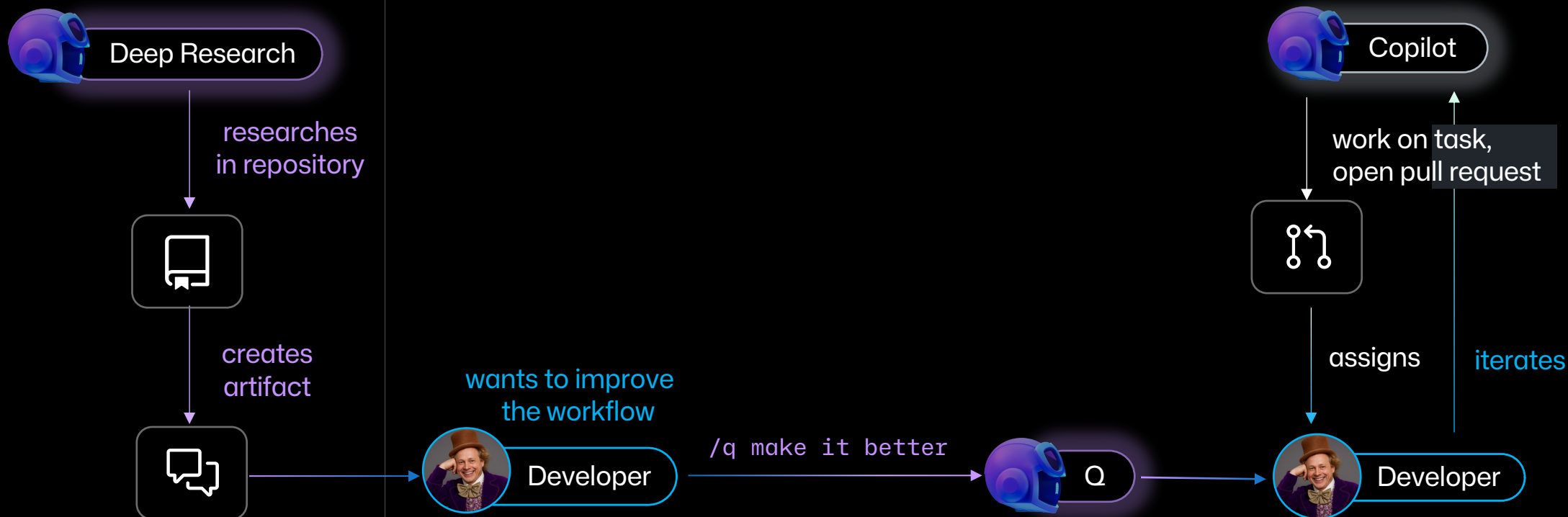


Q

The Agent Improver /q

RESEARCH

OPTIMIZE



At scale, agentic AI creates a new problem

The hidden token tax

- Costs accumulate quietly
Agentic AI jobs run automatically.
- Small inefficiencies compound
A few wasted calls per run become expensive.
- Raw token counts can mislead
Model choice, output tokens, workload size, etc.



Cost Management

to guard, measure, optimize

daily-pr-summary.md ✕

.github > workflows > daily-pr-summary.md > abc # Daily PR Summary

```
1 ---
2 on: daily
3 tools:
4   github: pull_requests
5 safe-outputs:
6   create-issue:
7 ---
8 # Daily PR Summary
9
10 Create an issue with the executive summary of the activity
11 in the repo yesterday.
```

daily-pr-summary.md ✕

daily-pr-summary.md > abc # Daily PR Summary > abc ## models: haiku

```
1 ---
2 on: weekly
3 tools:
4   github: pull_requests
5 safe-outputs:
6   create-issue:
7 steps: # deterministic compute
8   - run: gh pr list -after -1d > prs.json
9 max-ai-credits: 500
10 max-daily-ai-credits: 1000
11 ---
12 # Daily PR Summary
13
14 (1) Summarize each PR in prs.json with the `pr-summary` subagent
15 (2) Create executive summary from PR summaries into an issue
16
17 ## agent: `pr-summary`
18 ---
19 models: haiku
20 ---
21 Create an executive summary of the PR.
```

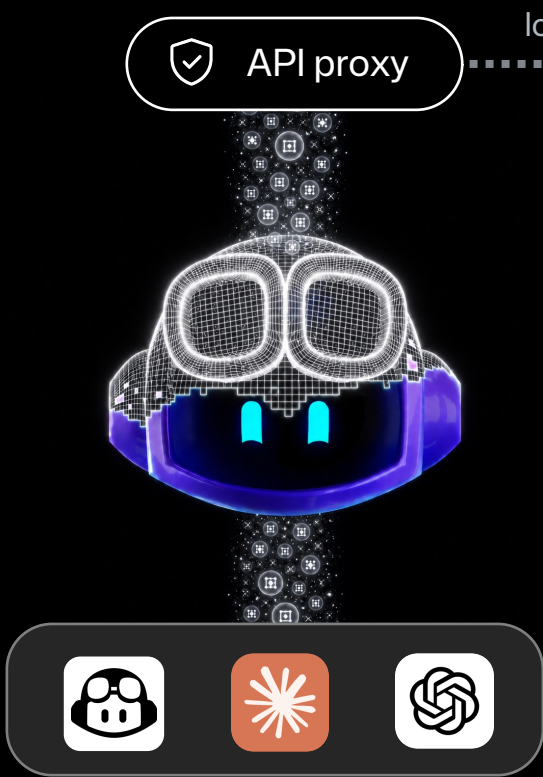
 Scheduling

 Deterministic

 Cost

 Sub-Agent

Make every token traceable



AI Credits = \$ 0.01 USD

#	Alias	Input	Output	Cache Read	Cache Write	ΔAI Credits	AI Credits	Duration
1	opus48	8,070	20	0	0	4.08	4.08	6.7s
2	opus48	1	564	0	9,361	7.26	11.3	6.7s
3	opus48	2	584	9,361	1,934	3.14	14.5	7.5s
...								
29	opus48	2	621	39,123	790	4	95.4	8.8s
30	opus48	2	318	39,913	752	3.26	98.7	4.5s
Total		8,127	11,630	803,564	40,665		94.7	2m58s

Table: Token usage for a “Smoke Claude” workflow run.

Make every token count



Quick wins

Remove obvious waste

- ☐ Prune unused MCP calls
- ☐ Stop repeated tool calls
- ☐ Fix fallback loops



Structural fixes

Move work out of the agent

- ☐ Pre-fetch diffs, files, metadata
- ☐ Add gates to skip irrelevant runs
- ☐ Use CLI proxies for deterministic reads



Strategic leverage

Optimize the system

- ☐ Right-size models by task
- ☐ Split monoliths into subagents
- ☐ Run A/B experiments
- ☐ Tune scheduling

Make optimization continuous



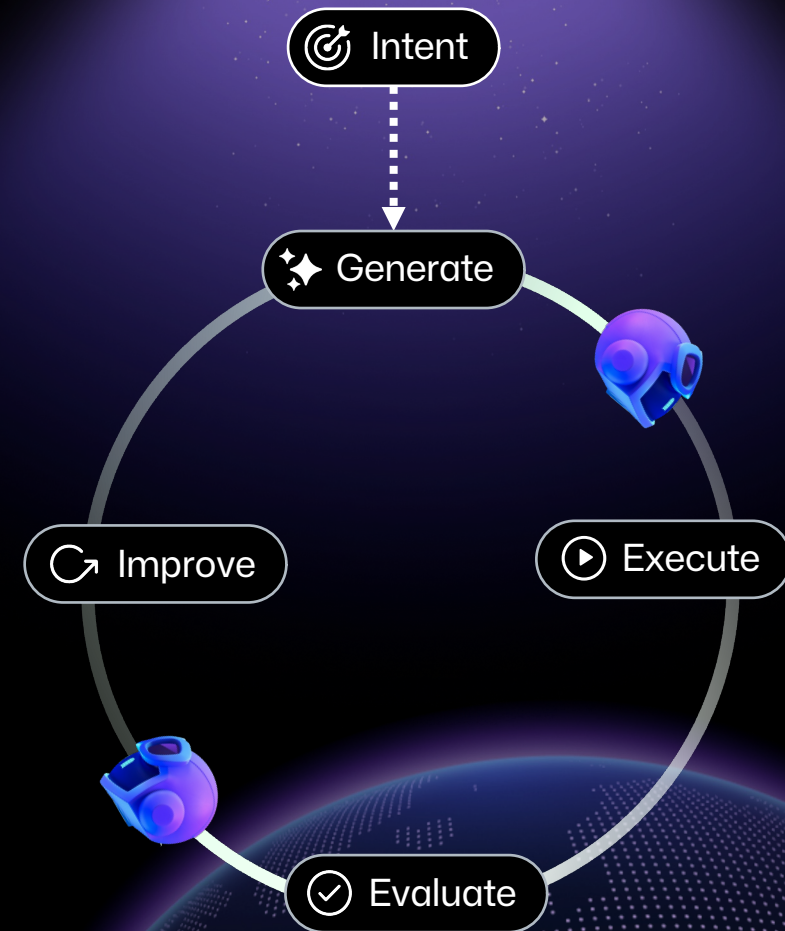
Install the GitHub Agentic Workflows CLI:

```
gh extensions install github/gh-aw
```



Add our automatic token audit & optimizers to your central org repository:

```
gh aw add githubnext/agent-ops
```





Agents are
power tools



Agentic workflows
are machine tools



Human Agent Processes are factories

Patterns

See our research for more useful design patterns.

gh.io/gh-aw

Welcome to Peli's Agent Factory

Jan 12, 2026



Don Syme



Peli de Halleux



Mara Kiefer

Welcome, welcome, WELCOME to Peli's Agent Factory!

Imagine a software repository where AI agents work alongside your team - not replacing developers, but handling the repetitive, time-consuming tasks that slow down collaboration and forward progress.

Peli's Agent Factory is our exploration of what happens when you take the design philosophy of **“let's create a new automated agentic workflow for that”** as the answer to almost every opportunity that arises! What happens when you **max out on automated agentic workflows** - when you make and use dozens of specialized, automated AI agentic workflows and use them in practice.



